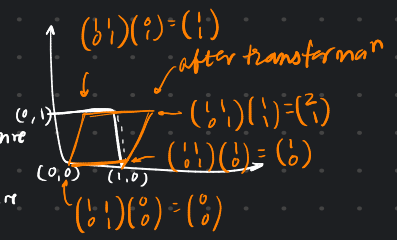
॥श्रीगणेशाय नमः॥

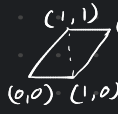
Determinants

- amount of change a matrix (↳ transformᵐ) can bring to a square of unit area, cube with unit volume, etc.

- E.g. $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ apply unit sq. (↳ a)



$$\det(a) = \frac{\text{area of new structure}}{\text{area of original structure}} = \frac{1}{1} = 1$$

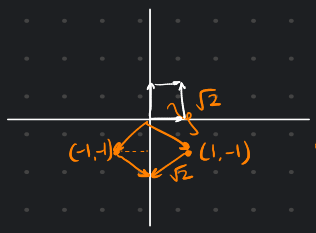


$A = 2 \times \frac{1}{2} \times 1 \times 1 = 1$

use of determinants
- checking invertibility:
  If $\det A = 0 \Rightarrow A$ is not invertible. (singular)
- checking whether a system of lin. eq^{n}s has 1 soln, ∞ soln's, or 0 soln.

e.g. 2.

$$m = \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix}$$



$A = \frac{1}{2} \times \sqrt{2} \times \sqrt{2} \times 2 = 2$

Determinant is the area/volume bounded by col^m vectors of the matrix.

If the transformᵃⁿ reduces the dimension of the original structure, then determinant of transformᵃⁿ is 0.

Say we have a rectangular matrix.

$$m = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}$$ this would take a 2-D fig (area) to 3-D fig. (vol^m), so we cannot compare.

Determinant of non-square transformaⁿs doesn't exist.

Properties of determinants

- $\det(I_{n\times n}) = 1$
- If we exchange a pair of rows or col^{m}s, the sign of determinant changes.

$$\det\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1 \qquad \det\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = -1$$

so even if of row or col^{m}s swaps would stay the same.

- Determinant is linear in terms of a single row.

Homogeneity:

For determinants:

$$2\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} 2a & 2b \\ c & d \end{vmatrix} = \begin{vmatrix} a & b \\ 2c & 2d \end{vmatrix}$$

$$\begin{vmatrix} ka & kb \\ c & d \end{vmatrix} = k\begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

$$\begin{vmatrix} ka & kb \\ kc & kd \end{vmatrix} = k^2\begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

(one from each row)

For matrices,

$$2\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 2a & 2b \\ 2c & 2d \end{bmatrix} \neq \begin{vmatrix} 2a & 2b \\ 2c & 2d \end{vmatrix}$$

For addition:

For matrices: $\begin{pmatrix} a & b \\ c & d \end{pmatrix} + \begin{pmatrix} x & y \\ z & w \end{pmatrix} = \begin{pmatrix} a+x & b+y \\ c+z & d+w \end{pmatrix}$

$m_1 \quad m_2 \quad m_3$

doesn't work for determinants. Have to do individually.

$|m_1| + |m_2| \neq |m_3|$

(can break row-wise)

$$\begin{vmatrix} a+b & c+d \\ e & f \end{vmatrix} = \begin{vmatrix} a & c \\ e & f \end{vmatrix} + \begin{vmatrix} b & d \\ e & f \end{vmatrix}$$

Diagonal matrix

$$\begin{bmatrix} d_1 & 0 & 0 & \cdots & 0 \\ 0 & d_2 & & & \vdots \\ 0 & & \ddots & & 0 \\ \vdots & & & & d_n \\ 0 & \cdots & 0 & \cdots & \end{bmatrix}$$

For diagonal matrix, $\det(A) = \prod_{i=1}^{n} A_{ii}$ ← product of diagonal entries.

$$\begin{vmatrix} d_1 & 0 & 0 \\ 0 & d_2 & \\ 0 & & d_n \end{vmatrix} = d_1 \begin{vmatrix} d_2 & 0 \\ 0 & d_n \end{vmatrix} = d_1 d_2 \begin{vmatrix} 1 & 0 \\ 0 & d_n \end{vmatrix} = d_1 \times d_2 \times \cdots \times d_n \underbrace{\begin{vmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{vmatrix}}_{\det(I_{n\times n}) = 1} = \prod_{i=1}^{n} d_i$$

Upper Δar matrix

↳ or lower

$$\begin{vmatrix} d_1 & a & b \\ 0 & d_2 & c \\ 0 & 0 & d_3 \end{vmatrix} \xrightarrow{\text{rref}} \begin{vmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{vmatrix} \qquad \det = d_1 \cdot d_2 \cdot d_3$$

$$= \begin{vmatrix} d_1 & 0 & 0 \\ 0 & d_2 & c \\ 0 & 0 & d_3 \end{vmatrix} + \underbrace{\begin{vmatrix} 0 & a & b \\ 0 & d_2 & c \\ 0 & 0 & d_3 \end{vmatrix}}_{0}$$

$$= \begin{vmatrix} d_1 & & \\ & d_2 & \\ & & d_3 \end{vmatrix} + \underbrace{\begin{vmatrix} d_1 & 0 & 0 \\ 0 & 0 & c \\ 0 & 0 & d_3 \end{vmatrix}}_{0}$$

$$= d_1 \cdot d_2 \cdot d_3$$

Det of upper or lower Δar matrix = Product of diagonal entries.

- Applying $R_i \leftarrow R_i \pm K \cdot R_j$ doesn't change the determinant.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \xleftarrow{R_1 \leftarrow R_1 - KR_2} \begin{pmatrix} a-kc & b-kd \\ c & d \end{pmatrix}$$

$$\det = \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \underbrace{\begin{vmatrix} -kc & -kd \\ c & d \end{vmatrix}}_{-k\begin{vmatrix} c & d \\ c & d \end{vmatrix} = -k\left[\begin{vmatrix} 0 & d \\ c & d \end{vmatrix} + \begin{vmatrix} c & 0 \\ c & d \end{vmatrix}\right]}$$

$$\underbrace{\begin{vmatrix} 0 & d \\ c & 0 \end{vmatrix}}_{-cd} \quad \underbrace{\begin{vmatrix} 0 & d \\ 0 & d \end{vmatrix}}_{0} \quad \underbrace{\begin{vmatrix} c & 0 \\ c & 0 \end{vmatrix}}_{0} \quad \underbrace{\begin{vmatrix} c & 0 \\ 0 & d \end{vmatrix}}_{cd} = 0$$

$\begin{vmatrix} c & 0 \\ c & 0 \end{vmatrix}$ → say det = d; then on swapping r_1 & r_2 $\begin{vmatrix} c & 0 \\ c & 0 \end{vmatrix}$ should be $-d$.

$\Rightarrow d = -d$

$\Rightarrow d = 0$.

- Matrix with repeated rows has determinant 0.

↳ or 11 rows $\begin{vmatrix} a & b \\ ka & kb \end{vmatrix}$

$$\begin{vmatrix} a & b & c \\ a & b & c \\ d & e & f \end{vmatrix} = d \qquad \text{swap 1st 2 rows} \qquad \begin{vmatrix} a & b & c \\ a & b & c \\ d & e & f \end{vmatrix} = -d \qquad \Rightarrow d = -d \qquad \Rightarrow d = 0$$

$$\text{Ily, } \begin{vmatrix} ka & kb & kc \\ a & b & c \\ d & e & f \end{vmatrix} = k \begin{vmatrix} a & b & c \\ a & b & c \\ d & e & f \end{vmatrix} = 0$$

- matrix with zero row has determinant 0
  (or col^m)

$$\begin{vmatrix} 0 & 0 \\ a & b \end{vmatrix} = \underbrace{\begin{vmatrix} 0 & 0 \\ a & 0 \end{vmatrix}}_{\text{lower bar}} + \underbrace{\begin{vmatrix} 0 & 0 \\ 0 & b \end{vmatrix}}_{\text{diagonal}} = 0 + 0 = 0$$

$\begin{vmatrix} a & 0 \\ b & 0 \end{vmatrix}$ ← lower bar $= 0$

Calculating determinants → cofactor formula, LU decomposition, Big formula

Cofactor formula

$$M = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

minor and cofactor are defined for each element

$$\text{minor}(a) = \det \begin{pmatrix} e & f \\ h & i \end{pmatrix} \qquad \text{minor}(b) = \begin{vmatrix} d & f \\ g & i \end{vmatrix}$$

minor of M_{ij}: determinant of submatrix formed by removing row i, col^m j.

$$\text{cofactor}(M_{ij}) = (-1)^{i+j} \cdot \text{Minor}(M_{ij})$$

For a 1×1 matrix, $\det[i] = i$

Determinant(M) = $a \cdot \text{cofactor}(a) + b \cdot \text{cofactor}(b) + c \cdot \text{cofactor}(c)$ } cofactor expansion to find det

$= a\,\text{cofactor}(a) + d \cdot \text{cofactor}(d) + g \cdot (\text{cofactor}(g)$

$= m_{11} C_{11} \cdot m_{12} C_{12} \cdots m_{1n} C_{1n}$

For 2×2 matrix:

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = a \cdot C_{11} + b \cdot C_{22} = a \cdot \left[(-1)^{1+1} |d|\right] + b\left[(-1)^{1+2} |c|\right] = ad - bc$$

$\det = ad - bc$

For 3×3 matrix:

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a \cdot (-1)^{1+1}(ei - hf) + b \cdot (-1)^{1+2}(di - gf) + c(-1)^{1+3}(dh - eg)$$

$$= a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

signs like $\begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix}$ ...

Big formula (use 3 properties, det(I) = 1, swap to get −ve det, split)

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} a & 0 \\ c & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & d \end{vmatrix}$$

$$\begin{vmatrix} a & 0 \\ c & 0 \end{vmatrix} \quad \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} \quad \begin{vmatrix} 0 & b \\ c & 0 \end{vmatrix} \quad \begin{vmatrix} 0 & b \\ 0 & d \end{vmatrix}$$

0 ; ad ; $-\begin{vmatrix} c & 0 \\ 0 & b \end{vmatrix}$; 0

$\det = ad - bc$

ad $-bc$

- $\det(A) = \det(A^T)$
- $\det(AB) = \det(A) \cdot \det(B)$

$\Rightarrow \det(A^{-1}) = \frac{1}{\det(A)}$

- $\det(kA) = k^n \det(A)$ (k: scalar, n×n matrix)

For easier determinant computation:
- find row/col^m with max. zeros
- try row operations to get upper/lower diagonal matrix
- use properties to simplify the matrix.

Q. Prove $\det(A) = \det(A^T)$.

→ $\det\begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$

$\det\begin{bmatrix} a & c \\ b & d \end{bmatrix} = ad - bc$

Ig for 3×3:

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a\det\begin{pmatrix} e & f \\ h & i \end{pmatrix} - b\cdot\det\begin{pmatrix} d & f \\ g & i \end{pmatrix} + c\cdot\det\begin{pmatrix} d & e \\ g & h \end{pmatrix}$$

$$\begin{vmatrix} a & d & g \\ b & e & h \\ c & f & i \end{vmatrix} = a\cdot\det\begin{pmatrix} e & h \\ f & i \end{pmatrix} - b\det\begin{pmatrix} d & g \\ f & i \end{pmatrix} + c\cdot\begin{pmatrix} d & g \\ e & h \end{pmatrix}$$

] same

Q. Take an $n \times n$ matrix with 1's above and below the diagonal, 0's on the diagonal.

e.g. $1\times1 \to \underset{C_1}{[0]}$, $2\times2 \to \underset{C_2}{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}}$, ...

Find the relation betn C_n, C_{n-1}, and C_{n-2}.

→ $[0] \to \det = 0$

$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \to \det = 0-(1) = -1$

$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \to \det = -1(-1) + 1(1) = 2$

$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}_{4\times4} \to \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix}$ $\det = 1\begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{vmatrix} + (-1)(-1)\begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} = -3$

$\begin{vmatrix} & \\ & \end{vmatrix}_{5\times5} = ?$

Det of $(n\times n)$ matrix $= (-1)^{(n-1)(n-1)}$

$\det(C_n) = [\text{abs}(\det(C_{n-1})) + 1] \times (-1) \times \dfrac{|\det(C_{n-2})|}{\det(C_{n-2})}$

Q. If the det of a matrix is 0, we cannot have unique soln to system of linear eqns.
Prove using elimination matrices.

→ If det = 0, then rref will have 1 zero row.

⇒ no soln or ∞ly many soln.

LU decomposition

$A = LU$

L: lower triangular matrix, $\det = 1$

U: upper triangular matrix, $\det = \det|A|$ = product of diagonal matrices

$u_{22} = \frac{|A_2|}{|A_1|}$

e.g. $A = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix} = LU = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 & 0 \\ 0 & \frac{2}{3} & 1 & 0 \\ 0 & 0 & -\frac{3}{4} & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 & 0 & 0 \\ & \frac{3}{2} & -1 & 0 \\ & & \frac{4}{3} & -1 \\ & & & \frac{5}{4} \end{bmatrix}$

$|A| = 2 \cdot \frac{3}{2} \cdot \frac{4}{3} \cdot \frac{5}{4} = 5$

$A = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$ matrix is kinda nice, symmetric, same diagonal elements, the result comes out same as well.

break into sub-matrices

$A_1 = [2]$

$A_2 = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = L \begin{bmatrix} 2 & -1 \\ 0 & \frac{3}{2} \end{bmatrix}$

$A_3 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$

$A_4 = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$

$|A_2| = 2 \cdot \frac{3}{2} = 3 = |A| \cdot u_{22}$

$|A_3| = 2 \cdot \frac{3}{2} \cdot \frac{4}{3} = 4 = |A_2| \cdot u_{33}$

Q. Take an $n \times n$ matrix with 1's above and below the diagonal, 0's on the diagonal.

e.g. $1\times1 \rightarrow [0]$, C_1; $2\times2 \rightarrow \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, C_2; ...

Find the relation betn C_n, C_{n-1}, and C_{n-2}.

$\rightarrow \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

$A_1 = [0]$ $\det = 0$

$A_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ $\det = -1$

$A_3 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ $\det = 0$

$A_4 \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ $\det = -1 \begin{vmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix} = (-1)\left[1(-1) + (-1)(0)\right] = 1$

$|A_n| = (-1)|A_{n-2}|$

Inverse of matrix using determinant

Adjoint :

create a cofactor matrix.

$$\begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix} \rightarrow \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix}$$

Take transpose

$$\begin{bmatrix} C_{11} & C_{21} \\ C_{12} & C_{22} \end{bmatrix} \leftarrow \text{Adjoint of } m$$

$$\text{Adjoint} = \left(\begin{matrix}\text{cofactor} \\ \text{matrix}\end{matrix}\right)^T$$

$$A^{-1} = \frac{1}{|A|} \cdot adj(A)$$

Cramer's rule

$$Ax = b \quad \Rightarrow \quad x = A^{-1} b$$

If A is full rank matrix, then $|A| \neq 0$.

$$A^{-1}\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = 2^{nd} \text{ col}^m \text{ of } A^{-1}$$

Acc. to cramer's rule :

$$x = \begin{bmatrix} |A_1| \\ |A_2| \\ |A_3| \end{bmatrix} \cdot \frac{1}{|A|}$$

$$|A_1| = \begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix} \quad |A_2| = \begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix} \quad |A_3| = \begin{vmatrix} a_{11} & a_{21} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{vmatrix}$$

Proof :

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = aC_a + bC_b + cC_c$$

$$\underbrace{aC_e + bC_d + cC_g}_{\text{what if we do this ?} \rightarrow 0}$$

(C ← cofactor matrix)

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} C_a & C_b & C_c \\ C_d & C_e & C_f \\ C_g & C_h & C_i \end{bmatrix}^T = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} C_a & C_d & C_g \\ C_b & C_e & C_h \\ C_c & C_f & C_i \end{bmatrix} = \begin{bmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{bmatrix} = |A| \cdot I$$

$$A \cdot C^T = |A| \cdot I$$

$$A^{-1} = \frac{C^T}{|A|}$$

$$x = A^{-1} b$$

$$= \frac{1}{|A|} C^T \cdot b$$

$$= \frac{1}{|A|} \begin{bmatrix} C_a & C_d & C_g \\ C_b & C_e & C_h \\ C_c & C_f & C_i \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \frac{1}{|A|} \cdot \begin{bmatrix} b_1 \cdot C_a + b_2 \cdot C_d + b_3 \cdot C_g \\ b_1 C_b + b_2 \cdot C_e + b_3 \cdot C_h \\ b_1 \cdot C_c + b_2 \cdot C_f + b_3 \cdot C_i \end{bmatrix}$$

det $\begin{vmatrix} b_1 & b & c \\ b_2 & e & f \\ b_3 & h & i \end{vmatrix}$